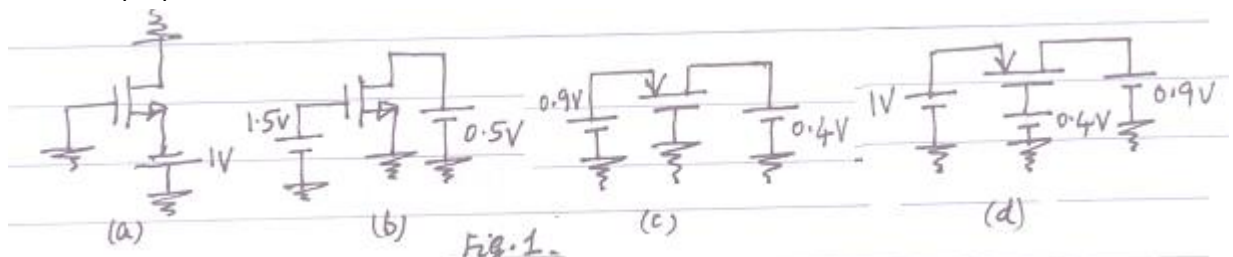


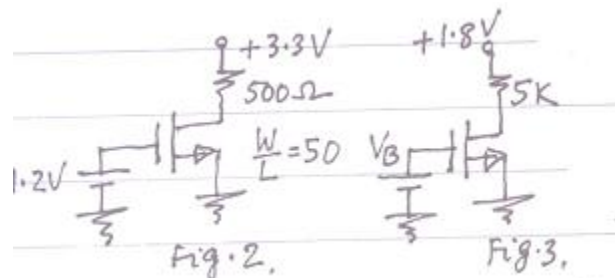
Department of Electronics and Communications Engineering
3rd Semester- Analog Microelectronic Circuits (21EC33)
Tutorial-1 (Unit 2 and Unit 3)

03-01-2023

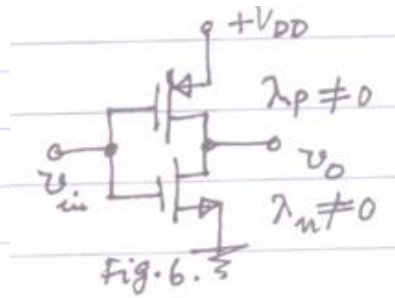
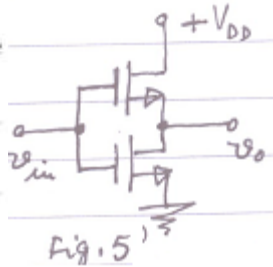
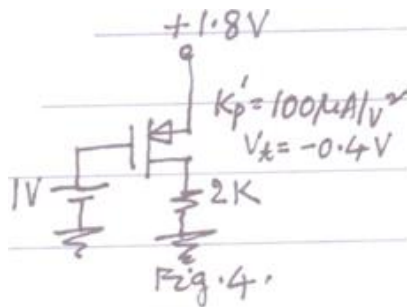
1. In a process technology for which $t_{ox} = 12\text{nm}$ and $\mu_n = 600\text{cm}^2/\text{v}.\text{sec}$, find C_{ox} and k_n' and the over drive voltage V_{OV} required for an NMOS transistor, with an aspect ratio of 15, to operate in saturation with a drain current $= 150\mu\text{A}$. Determine the minimum value of V_{DS} needed. Also calculate r_{DS} for small values of V_{DS} .
2. A PMOS transistor has $t_{ox} = 20\text{nm}$, $\mu_p = 400\text{cm}^2/\text{v}.\text{sec}$, $V_t = -0.8\text{V}$ and $W/L = 12$. Find the drain current for the following cases. a) $V_{GS} = -5\text{V}$ & $V_{DS} = -1\text{V}$
b) $V_{GS} = -2\text{V}$ & $V_{DS} = -1.2\text{V}$.
3. An NMOS transistor operating in linear region with $V_{DS} = 0.1\text{V}$ is found to conduct $60\mu\text{A}$ with $V_{GS} = 2\text{V}$ and $160\mu\text{A}$ with $V_{GS} = 4\text{V}$. Find V_t of the device.
4. Determine the region of operation of the MOSFET in the circuits of fig1. Assume $|V_t| = 0.4\text{V}$.



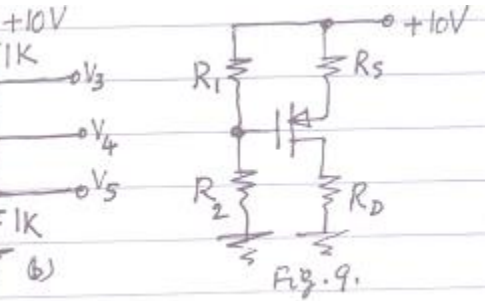
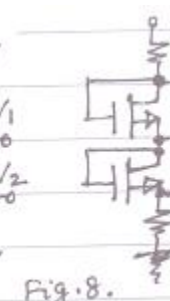
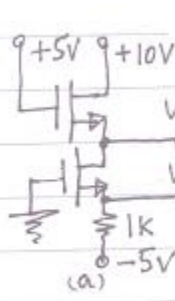
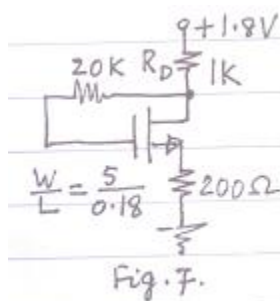
5. In the circuit of fig2, calculate the minimum allowable V_{DD} , so that the transistor must not enter triode region. Assume $V_t = 0.4\text{V}$, $k_n' = 200\mu\text{A}/\text{v}^2$ and $\lambda = 0$.



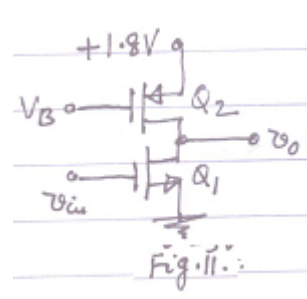
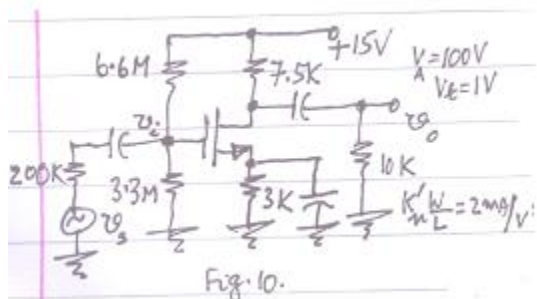
6. The NMOS transistor in fig 3 has $V_t = 0.5\text{V}$, aspect ratio $= 200$, $k_n' = 0.2\text{mA}/\text{v}^2$. What value of V_B places the transistor at the edge of saturation?
7. An NMOS device operating in saturation with $\lambda = 0$ must provide a trans-conductance of $1/50\Omega$. $V_t = 0.4\text{V}$ and $k_n' = 0.2\text{mA}/\text{v}^2$. a) Determine W/L if $I_D = 0.5\text{mA}$. b) Determine W/L if $V_{GS} - V_t = 0.5\text{V}$. c) Determine I_D if $V_{GS} - V_t = 0.5\text{V}$.
8. An NMOS device with $V_A = 10\text{V}$ must provide an intrinsic gain of 20 with $V_{DS} = 1.5\text{V}$. Determine the required value of W/L , if $I_D = 0.5\text{mA}$. Assume $V_t = 0.4\text{V}$ and $k_n' = 0.2\text{mA}/\text{v}^2$.
9. If $\lambda = 0$, what value of W/L places the P-channel device at the edge of saturation in the circuit of fig4?



10. Draw the small signal equivalent model for the circuit in fig5. Assume all transistors operate in saturation and $\lambda \neq 0$.
11. For the circuit of fig6, draw the small signal equivalent model and determine the small signal voltage gain, v_o/v_i . Assume $\lambda_n \neq 0$ and $\lambda_p \neq 0$.
12. Determine the drain current in the circuit of fig7, if $\mu_n C_{ox} = 100 \mu A/V^2$, $V_t = 0.5V$ and $\lambda = 0$. What value of R_D is necessary to reduce I_D by a factor of 2?
13. For the two circuits shown in fig8, determine V_1, V_2, V_3, V_4 and V_5 . $K_n' W/L = 2mA/V^2$, $V_t = 1V$ and $\lambda = 0$.

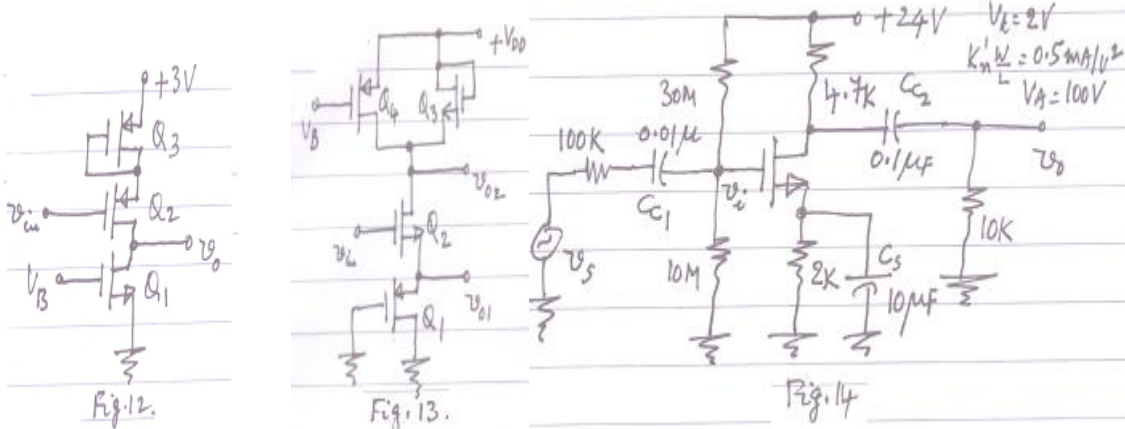


14. Design the circuit in fig9, so that the PMOS device operates in saturation with V_D biased 1V from the edge of triode region, with $I_D = 1mA$ and $V_D = 3V$. The device has $V_t = -1V$ and $k_p' W/L = 0.25mA/V^2$. Assume $10 \mu A$ current in the voltage divider.
15. In the common source amplifier circuit of fig10, determine I_D, g_m and r_o . Draw the small signal equivalent circuit and calculate $R_{in}, R_o, v_i/v_s, v_o/v_i$ and v_o/v_s .



16. The power budget of the CS amplifier circuit with an active load shown in fig11 is $0.9mW$. Determine the aspect ratio of Q_1 , if $v_o/v_i = -10$. $K_n' = 0.2mA/V^2$, $k_p' = 0.1mA/V^2$, $\lambda_1 = 0.1V^{-1}$ and $\lambda_2 = 0.15V^{-1}$. If $(W/L)_2 = 100$, determine V_B . $|V_t| = 0.4V$.

17. Calculate the voltage gain v_o/v_i in the circuit of fig12. The transistors are biased such that the drain current is 0.5mA. $g_{m3} = 2g_{m2} = 4g_{m1} = 4\text{mA/V}$ and $V_{A3}=1.25\text{V}$ $V_{A2} = 2V_{A1} = 20\text{V}$.
18. The transistors in the circuit of fig13 are biased such that $I_{D1} = 1\text{mA}$ and $I_{D4} = 0.25\text{mA}$. Find v_{o1}/v_i and v_{o2}/v_i . $g_{m4} = g_{m3} = 0.5g_{m2} = 0.25g_{m1} = 1\text{mA/V}$, $\lambda_1 = \lambda_3 = \lambda_4 = 0.05\text{V}^{-1}$ and $\lambda_2 = 0$.



19. In the RC coupled CS amplifier circuit of fig14, determine I_D , g_m and the mid frequency gain, A_M of the amplifier. Determine the cut off frequencies f_{L1} , f_{L2} and f_{L5} due to C_{C1} , C_{C2} and C_S respectively. If $C_{gs} = 1\text{pF}$ and $C_{gd} = 0.2\text{pF}$, calculate the unity gain frequency of the transistor. Determine the upper cutoff frequencies f_{Hi} and f_{Ho} due the input and output capacitances respectively. Also obtain the 3dB band width of the amplifier.
